

$\mathbb{Q}$ 的加法和乘法

Lemma: 对 $\forall s_1, s_2 \in \mathbb{Z} \setminus \{0\}$, 有: $[s_1, s_1] = [s_2, s_2]$

Proof: $\because s_1, s_2 \in \mathbb{Z} \setminus \{0\} \quad \therefore s_1 s_2 = s_2 s_1$
 $\downarrow$
 $\mathbb{Z}$ 中的乘法交换律

$$\therefore (s_1, s_1) \sim (s_2, s_2) \quad \therefore [s_1, s_1] = [s_2, s_2] \quad \square$$

定义 ($\mathbb{Q}$ 中的乘法幺元) 任取 $s \in \mathbb{Z} \setminus \{0\}$, 定义 $1 := [s, s]$

Remark: 上一引理保证了这一定义的良好性.

Lemma ($\mathbb{Q}$ 中 1 的乘法性质) 对 $\forall x \in \mathbb{Q}$, 有: $x \cdot 1 = x = 1 \cdot x$

Proof: $\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

任取 $\alpha \in \mathbb{Z} \setminus \{0\} \quad \therefore 1 = [\alpha, \alpha]$

$$\therefore x \cdot 1 = [r, s] \cdot [\alpha, \alpha] = [r\alpha, s\alpha] = [r, s] = x$$

$$1 \cdot x = [\alpha, \alpha] \cdot [r, s] = [\alpha r, \alpha s] = [r\alpha, s\alpha] = [r, s] = x$$

$$\therefore x \cdot 1 = x = 1 \cdot x \quad \square$$

Lemma: $\lambda \in \mathbb{Q}$ 若对 $\forall x \in \mathbb{Q}$, 有: $x \cdot \lambda = x = \lambda \cdot x$, 则有: $\lambda = 1$

Proof: $\because \lambda \in \mathbb{Q} \quad \therefore \exists \gamma \in \mathbb{Z}, \delta \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } \lambda = [\gamma, \delta]$

$\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

$$\therefore x \cdot \lambda = x \quad \therefore [r, s] \cdot [\gamma, \delta] = [r, s]$$

$$\therefore [r\gamma, s\delta] = [r, s] \quad \therefore (r\gamma, s\delta) \sim (r, s)$$

$$\therefore (r\gamma)s = r(sd) \quad \therefore (r\gamma)s = (rd')s$$

$$\therefore (r\gamma)s = (rd')s, s \in \mathbb{Z} \setminus \{0\} \quad \therefore r\gamma = rd'$$

$$\therefore \forall x \in \mathbb{Q}, \text{有: } x \cdot \lambda = x \quad \therefore \text{当 } r \neq 0 \text{ 时, } r\gamma = rd' \text{ 成立.}$$

$$\therefore \gamma = d'$$

$$\therefore \lambda = [\gamma, d'] = [d', d'] = 1 \quad \square$$

定义: 记 $\mathbb{Q}^\times := \mathbb{Q} \setminus \{0\}$.

Lemma: 对 $\forall x \in \mathbb{Q}^\times$, 存在唯一的 $x^{-1} \in \mathbb{Q}^\times$ 使得 $x \cdot x^{-1} = 1 = x^{-1} \cdot x$

Proof: $\because x \in \mathbb{Q} \quad \therefore \exists r \in \mathbb{Z}, s \in \mathbb{Z} \setminus \{0\}, \text{ s.t. } x = [r, s]$

假设 $r=0$, 则有: $x = [r, s] = [0, s] = 0 \quad \therefore 0 = x \in \mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ 矛盾.

$\therefore r \neq 0 \quad \therefore r \in \mathbb{Z} \setminus \{0\}$

$\therefore s \in \mathbb{Z} \setminus \{0\}, r \in \mathbb{Z} \setminus \{0\} \quad \therefore [s, r] \in \mathbb{Q} \quad \text{且 } rs \in \mathbb{Z} \setminus \{0\}$

假设 $[s, r] = 0$ 则有: $[s, r] = 0 = [0, s]$

$\therefore (s, r) \sim (0, s) \quad \therefore s \cdot s = 0 \cdot r \quad \therefore s \cdot s = 0 \quad \therefore s = 0$ 矛盾.

$\therefore [s, r] \neq 0 \quad \therefore [s, r] \in \mathbb{Q} \setminus \{0\} = \mathbb{Q}^\times \quad \text{令 } x^{-1} = [s, r] \in \mathbb{Q}^\times$

$$\therefore x \cdot x^{-1} = [r, s] \cdot [s, r] = [rs, sr] = [rs, rs] = 1$$

$$x^{-1} \cdot x = [s, r] \cdot [r, s] = [sr, rs] = [rs, rs] = 1$$

$\therefore x \cdot x^{-1} = 1 = x^{-1} \cdot x$ 存在性得证.

下证唯一性. 假设存在 $\alpha, \beta \in \mathbb{Q}^\times$, s.t. $x \cdot \alpha = 1 = \alpha \cdot x, x \cdot \beta = 1 = \beta \cdot x$

$$\text{则有: } \alpha = \alpha \cdot 1 = \alpha \cdot (x \cdot \beta) = (\alpha \cdot x) \cdot \beta = 1 \cdot \beta = \beta.$$

$\therefore$ 唯一性得证. $\square$

Lemma: 对 $\forall x \in \mathbb{Q}^x$, 有: $(x^{-1})^{-1} = x$

Proof: $\because x \in \mathbb{Q}^x \therefore \exists$ 唯一的 $x^{-1} \in \mathbb{Q}^x$, s.t. $x \cdot x^{-1} = 1 = x^{-1} \cdot x$

$\because x^{-1} \in \mathbb{Q}^x \therefore \exists$ 唯一的 $(x^{-1})^{-1} \in \mathbb{Q}^x$, s.t. $x^{-1} \cdot (x^{-1})^{-1} = 1 = (x^{-1})^{-1} \cdot x^{-1}$

$$\therefore x^{-1} \cdot x = 1 = x \cdot x^{-1}$$

$$x^{-1} \cdot (x^{-1})^{-1} = 1 = (x^{-1})^{-1} \cdot x^{-1}$$

$$\therefore (x^{-1})^{-1} = x \quad \square$$

~~Lemma: 对 $\forall x, y \in \mathbb{Q}^x$, 有: $(xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1}$~~

Lemma ($\mathbb{Q}$ 满足加法结合律) 对 $\forall x, y, z \in \mathbb{Q}$, 有: $x + (y + z) = (x + y) + z$

Proof: 对 $\forall [m, n], [r, s], [p, q] \in \mathbb{Q}$, 有:

$$([m, n] + [r, s]) + [p, q] = [ms + rn, ns] + [p, q]$$

$$= [(ms + rn)q + p(ns), (ns)q] = [msq + rnq + pns, nsq]$$

$$= [m(sq) + (rq + ps)n, n(sq)] = [m, n] + [rq + ps, sq]$$

$$= [m, n] + ([r, s] + [p, q]) \quad \square$$

Lemma ($\mathbb{Q}$ 满足加法消去律) 对 $\forall x, y, z \in \mathbb{Q}$, 若 $x+z=y+z$, 则有 $x=y$

Proof: 对 $\forall [m, n], [r, s], [p, q] \in \mathbb{Q}$, ~~若~~

若 $[m, n] + [p, q] = [r, s] + [p, q]$, 则有: $[mq+pn, nq] = [rq+ps, sq]$

$$\therefore (mq+pn, nq) \sim (rq+ps, sq) \quad \therefore (mq+pn)(sq) = (rq+ps)(nq)$$

$$\therefore msq^2 + nspq = rnq^2 + nspq \quad \therefore msq^2 = rnq^2$$

$$\therefore q \in \mathbb{Z} \setminus \{0\} \quad \therefore q^2 \in \mathbb{Z} \setminus \{0\} \quad \therefore ms = rn \quad \therefore (m, n) \sim (r, s)$$

$$\therefore [m, n] = [r, s] \quad \square$$

Lemma ($\mathbb{Q}$ 满足乘法对非零元的消去律) 对 $\forall x, y, z \in \mathbb{Q}$, 若 $xz = yz$ 且 $z \neq 0$, 则有: $x=y$

Proof: 对 $\forall [m, n], [r, s], [p, q] \in \mathbb{Q}$,

若 $[m, n] \cdot [p, q] = [r, s] \cdot [p, q]$ 且 $[p, q] \neq 0$, 则有:

$$[mp, nq] = [rp, sq] \quad \therefore (mp, nq) \sim (rp, sq)$$

$$\therefore (mp)(sq) = (rp)(nq) \quad \therefore (msp)q = (rnp)q \quad \therefore q \in \mathbb{Z} \setminus \{0\}$$

$$\therefore (ms)p = (rn)p \quad \text{假设 } p=0, \text{ 则有: } [p, q] = [0, q] = 0. \text{ 矛盾. } \therefore p \neq 0$$

$$\therefore ms = rn \quad \therefore (m, n) \sim (r, s) \quad \therefore [m, n] = [r, s] \quad \square$$

Lemma: $\nexists \forall x, y \in \mathbb{Q}^x$, 有: $xy \in \mathbb{Q}^x$

Proof: $\because x, y \in \mathbb{Q}^x \quad \therefore x, y \in \mathbb{Q} \quad \therefore x \cdot y \in \mathbb{Q}$

$\because x, y \in \mathbb{Q}^x \quad \therefore x \neq 0 \text{ 且 } y \neq 0 \quad \therefore xy \neq 0$

$\therefore xy \in \mathbb{Q}^x \quad \square$

Lemma ($\mathbb{Q}$ 中无零因子) 对 $\forall x, y \in \mathbb{Q}$, 若 $x \neq 0$ 且 $y \neq 0$, 则有: $xy \neq 0$.

Proof: 假设 $xy = 0$. 则有: $xy = 0 = 0 \cdot y$

$\therefore x, 0, y \in \mathbb{Q}$, $xy = 0y$ 且 $y \neq 0 \quad \therefore x = 0$ 矛盾.

$\therefore xy \neq 0$. $\square$

Lemma: 对 $\forall x, y \in \mathbb{Q}$, 若 $xy = 0$, 则有: $x = 0$ 或 $y = 0$.

Proof: 由上引理, 显然. $\square$

Lemma ($\mathbb{Q}$ 满足乘法交换律) 对 $\forall x, y \in \mathbb{Q}$, 有: $xy = yx$.

Proof: 对 $\forall [m, n], [r, s] \in \mathbb{Q}$, 有:

$$[m, n] \cdot [r, s] = [mr, ns] = [rm, sn] = [r, s] \cdot [m, n] \quad \square$$

Lemma: 对 $\forall x, y \in \mathbb{Q}^x$, 有: $(xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1}$

Proof: $\therefore x, y \in \mathbb{Q}^x \quad \therefore x, y \in \mathbb{Q} \quad \therefore x, y \in \mathbb{Q}^x \quad \therefore x \neq 0$ 且 $y \neq 0$

$\therefore xy \neq 0 \quad \therefore xy \in \mathbb{Q}^x$

$\therefore x \in \mathbb{Q}^x \quad \therefore \exists$ 唯一的 $x^{-1} \in \mathbb{Q}^x$, s.t. $x \cdot x^{-1} = 1 = x^{-1} \cdot x$

$\therefore y \in \mathbb{Q}^x \quad \therefore \exists$ 唯一的 $y^{-1} \in \mathbb{Q}^x$, s.t. $y \cdot y^{-1} = 1 = y^{-1} \cdot y$

$\therefore xy \in \mathbb{Q}^x \quad \therefore \exists$ 唯一的 $(xy)^{-1} \in \mathbb{Q}^x$, s.t. $(xy) \cdot (xy)^{-1} = 1 = (xy)^{-1} \cdot (xy)$

$$\because y^{-1} \in \mathbb{Q}^x, x^{-1} \in \mathbb{Q}^x \quad \therefore y^{-1}x^{-1} \in \mathbb{Q}^x$$

$$\because (xy)(y^{-1}x^{-1}) = ((xy)y^{-1})x^{-1} = (x(yy^{-1}))x^{-1} = (x \cdot 1)x^{-1} = xx^{-1} = 1$$

$$(y^{-1}x^{-1})(xy) = ((y^{-1}x^{-1})x)y = (y^{-1}(x^{-1}x))y = (y^{-1} \cdot 1)y = y^{-1}y = 1$$

$$\therefore (xy)(y^{-1}x^{-1}) = 1 = (y^{-1}x^{-1})(xy)$$

$$(xy)(xy)^{-1} = 1 = (xy)^{-1}(xy)$$

$$\therefore (xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1} \quad \square$$